COMP7507 Visualization and Visual Analytics

Project Proposal

Important Notes:

- The purpose of the proposal is to help you define the project scope and consolidate your ideas. Try your best to explore and investigate the feasibility of your ideas & plans when developing your proposal. You should better identify viable data sources so that it does not take you too much time on tedious data cleaning or processing.
- You need to write the proposal (i.e., fill in the text boxes) **using this template** and **submit one for each group**. Your inputs should be in font **Time New Roman 12pt black/auto.** Part B should contain no more than **5 pages**.
- You will receive feedback on your proposal from the teaching team within 1 week after submission, but you should continue to work on the project after proposal submission without idling.

Part A *(Basic Information)*

(i) Group Information

Group Number: 14

Group Members: 5

Name	*Student ID*
Member A: Qin Yuhong	3036807519
Member B: Wang Chenyu	3036793770
Member C: Su Chuanzhou	3036837980
Member D: Cheng Jionghao	3036806711
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Project Title

Everest Climbing Decision Big Data Support: History, Routes, and Profile-Based Summit-Success Estimation

Part B *(Proposed Method) - Maximum 5 pages*

Project Objectives

We will develop an interactive visualization for people planning an Everest expedition. It will connect a historical view of expedition change, a route and terrain explorer, and a profile based planning tool. Users can compare seasons and routes, inspect geographic evidence, and receive an interpretable summit success estimate with uncertainty.

Dataset Description

We will use Maven Analytics' Himalayan Expeditions CSV dataset [2], based on The Himalayan Database [1]. The local snapshot contains 11,425 expedition records, approximately 89,000 member records, 480 peaks, and 15,586 references. Everest filtering gives 2,347 expeditions and 25,274 participation records from 1921 to 2024. Available fields include year, season, host country, route names, summit outcome, expedition duration, team size, hired support, oxygen use, member success, and recorded deaths. The dataset and data dictionary are stored in the project's data folder.

The analysis will distinguish expedition records, participation records, and unique people. Participation rows are not unique climbers, and the member identifier is expedition-local. The expedition identifier also contains a two-digit year, so records will be linked with a composite key such as peak, year, and expedition identifier after duplicate checks. Success, death, and risk rates will always state their numerator, denominator, unit, and sample size.

The local data contains route names but no complete route geometry, camp coordinates, or terrain model. We will add a separately cited route waypoint layer only where coordinates can be verified, and use NASA SRTM elevation data [9] for terrain context. The official Himalayan Database season lists [4] and Nepal Department of Tourism reports, including the 2026 spring aggregate reported by Radio Nepal [5] and Kathmandu Post [6], may provide a clearly labelled current snapshot. These aggregate records will not be merged into historical member-level modelling data.

Methodology

Tools and product flow. Python will clean and aggregate the CSV files, check keys and denominators, and build model inputs. D3 [8] will support linked analytical charts. MapLibre GL JS [7] will provide interactive 2D maps and terrain-aware 3D context. The offline application will be accessed through index.html. The main path is: historical evolution -> route and current-situation explorer -> profile-based historical estimate.

1. Module 1: historical evolution. Annual and decade trends will show expedition counts, participation records, seasonal composition, duration, team/support structure, oxygen use, summit success, and recorded deaths. A season-by-decade heatmap and linked interval charts will encourage users to compare patterns rather than read isolated totals. All views will distinguish counts from rates and show sample sizes.

2. Module 2: route and current-situation explorer. This is the primary exploration view. Users will filter by climbing side, route group, season, and period, then compare typical duration, team size, support, oxygen use, summit success, and recorded risk. A route panel will connect charts to a 2D map and a 3D terrain view containing verified camps, waypoints, elevation, and contextual annotations. If route geometry cannot be verified, the interface will use a clearly labelled schematic layer. Current 2025-2026 aggregate indicators will remain separate from the historical microdata.

3. Module 3: personalised plan and historical estimate. Users may enter an age band, recorded sex, target season, climbing side, route group, team size, support, and planned oxygen use. The system will rank comparable historical plans and estimate recorded expedition summit success. A regularized logistic regression will be compared with constant and side-by-season baselines. Only variables available before an expedition will be used as predictors; actual summit dates, reached altitude, actual deaths, and other post-expedition fields will be excluded. Results will include

comparable sample size, uncertainty intervals, calibration information, and the main explanatory variables. Sparse profiles will return insufficient evidence rather than an overconfident estimate.

4. Validation and visual analysis. Data checks will verify composite keys, filters, missingness, denominators, map interactions, empty states, and browser-versus-Python results. Training, validation, and final testing will use 1990-2014, 2015-2019, and 2020-2024, keeping each expedition in one split. We will compare Brier score, log loss, AUC, and calibration curves [10]. Visual analysis will ask whether seasonal and route differences remain after comparing similar groups. Results will be interpreted as associations in historical records, not causal or guaranteed individual effects.

Evaluation. Approximately six external users will complete tasks that follow the product path: identify a historical change, compare two routes, inspect the terrain view, enter a profile, and interpret the estimate and its limitations. Completion, time, and misunderstandings will guide revisions. If 3D geometry or the estimator cannot be validated by the integration deadline, the fallback is a stable 2D route explorer and a historical profile comparison without a numerical estimate.

Part C *(List of Tasks)*

Task List with Timeline and Division of Labour

Each member will contribute approximately 60 hours (300 hours total), including analysis, visualization, testing, and a report section. Work is organized around overlapping pairs so that every module has a primary owner and a review partner.

Member A — Primary data and provenance: clean files, define composite keys and cohorts, verify source versions, and maintain the data dictionary. A+B will release the historical analysis tables and denominator checks; A+E will document sources, file versions, and packaging.

Member B — Primary historical evolution: implement trends, season-by-period comparisons, interval charts, and linked filters. A+B will validate counts; B+C will convert route and side metrics into the comparison views used by Module 2.

Member C — Primary route and map explorer: curate verified waypoints, build the 2D/3D terrain context, and implement route comparison interactions. B+C will agree on route metrics; C+D will define the profile-plan features passed to Module 3.

Member D — Primary profile estimator: build baselines and the calibrated logistic model, audit pre-expedition features, and produce explanation charts. C+D will test route/profile feature mappings; D+E will integrate the estimate and uncertainty display.

Member E — Primary integration and evaluation: connect the three modules, build the user path and profile form, conduct task-based testing, and package the offline application. D+E will verify browser/model consistency; A+E will complete provenance, QA, and the final index.html. A-D remain responsible for their own analytical components.

9-10 October: finalize and submit the proposal. 11-16 October: A+B freeze the data contract and historical measures; C validates map-source feasibility; D defines model targets; E defines shared interfaces. 17-30 October: B completes Module 1; B+C build route metrics and Module 2; C prepares the map layer; D prepares model baselines; E integrates the first user path. 31 October-7 November: complete the route explorer and 3D/2D fallback, freeze model features, and perform key validation. 8-13 November: build Module 3, calibrate estimates, and connect explanations to evidence. 14-20 November: conduct user tasks, revise the interface, verify offline operation, and rehearse. 21 or 28 November: live demonstration according to course allocation. 30 November: submit the work ZIP, index.html, and combined reports; complete critique and peer evaluation by 5 December.

Priority order is Module 2 route exploration, Module 3 historical estimation, and Module 1 historical context. If time or data quality is limited, reduce decorative 3D detail, external current indicators, and numerical prediction before removing route comparisons, denominators, uncertainty, or explanatory evidence.

Part D *(References)*

[1] The Himalayan Database. (n.d.). The Himalayan Database: The expedition archives of Elizabeth Hawley. https://www.himalayandatabase.com/

[2] Maven Analytics. (n.d.). Himalayan Expeditions [Data set]. https://mavenanalytics.io/data-playground/himalayan-expeditions

[3] The Himalayan Database. (n.d.). Downloads and updates. https://himalayandatabase.org/downloads.html

[4] The Himalayan Database. (n.d.). Season lists. https://himalayandatabase.org/seasonlists.html

[5] Radio Nepal. (2026, June 10). Spring climbing season concludes; Nepal earns Rs 1.26 billion in revenue. https://radionepalonline.com/en/2026/06/10/431063.html

[6] Kathmandu Post. (2026, May 29). Everest spring climbing season sets new record with 1,008 successful summits. https://kathmandupost.com/national/2026/05/29/everest-spring-climbing-season-sets-new-record-with-1-008-successful-summits

[7] MapLibre. (n.d.). MapLibre GL JS documentation. https://maplibre.org/maplibre-gl-js/docs/

[8] D3. (n.d.). D3 documentation. https://d3js.org/

[9] NASA Earthdata Land Processes Distributed Active Archive Center. (n.d.). Shuttle Radar Topography Mission (SRTM). https://www.earthdata.nasa.gov/centers/lp-daac

[10] Scikit-learn developers. (n.d.). Probability calibration. https://scikit-learn.org/stable/modules/calibration.html

Web sources [1]-[10] accessed 9 October 2026. The analysis will use the local CSV snapshot with source versions and file checksums recorded for reproducibility.